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bisonpicsuite: A set of R packages to estimate wood bison population parameters from remote camera data.

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Software

- Review
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Statement of Need

Knowledge of population status and trends is integral to the effective conservation and management of wildlife populations. To this end, demographic ratios (e.g., Calf:Cow ratios, Fig. 1) are routinely used in wildlife management as a readily observable measure of productivity and for forecasting population trajectories (Bender, 2006; Fuller et al., 2007; Wittmer et al., 2005). More complex state-space population modeling approaches (Buckland et al., 2004; Mizuki et al., 2020; Newman et al., 2023; Paterson et al., 2019) can also be used to estimate survival and fecundity. To understand the age composition of a population, both approaches require classified (by age and sex) counts of individuals.

In remote areas, estimates of herd size and composition are typically obtained from aerial surveys. However, distinguishing animal age and sex can be challenging, particularly in forested environments if animals flee or hide from aircraft. Remote cameras present an alternative method for obtaining classified counts. Wildlife cameras have been used to estimate species occupancy, density, behaviour, and to identify individuals through distinct markings (Caravaggi et al., 2017; Green et al., 2020; Magoun et al., 2011; Nakashima et al., 2020; Singh et al., 2022; Steenweg et al., 2016). Although different age and sex classes can be readily distinguished in remote camera photos for many ungulate species (Laskin et al., 2020), classified count camera data are seldom utilized to estimate population parameters without using individual markings (but see Horne & Thompson (2021), Chitwood et al. (2017), Ikeda et al. (2013)).

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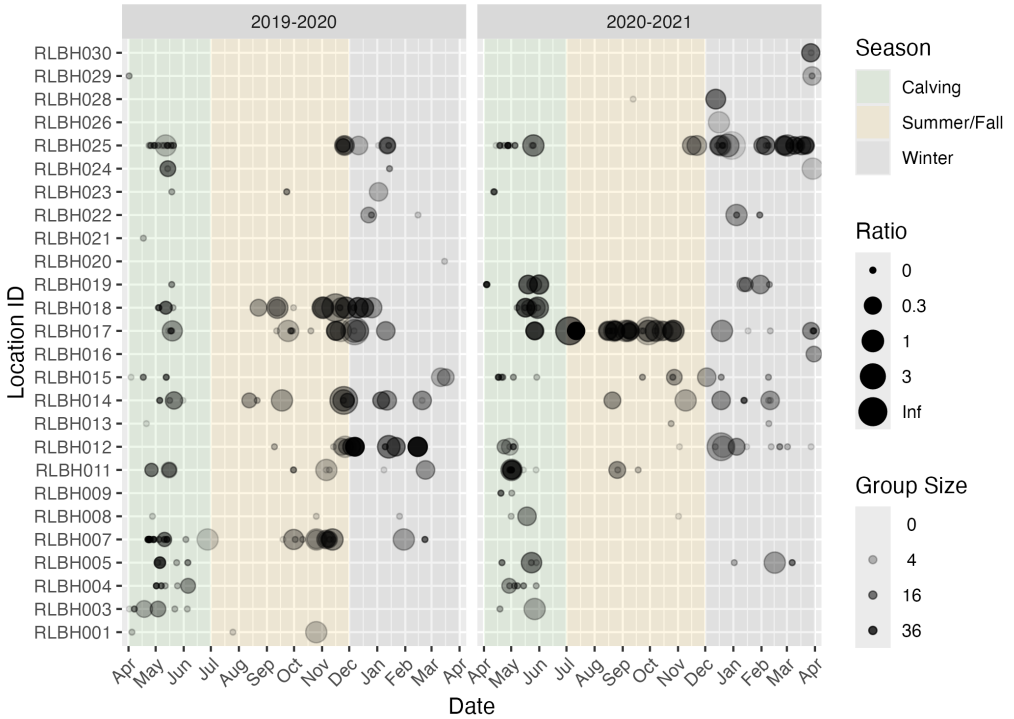

Figure 1: *Figure 1. Data exploration plot from the shinybisonpic app, showing calf:cow ratios over time from individual remote cameras in the Ronald Lake Wood Bison range in northeast Alberta. Ratios are shown with camera trap ID, observation date, study year, season, and group size. A ratio of 0 represents a group of entirely cows, while an infinite ratio (Inf) represents a group of entirely calves.*

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Summary

We present a method utilizing classified counts from remote cameras to evaluate wood bison (*Bison bison athabascae*) herd demographics. We modeled demographic ratios, survival, and productivity using a Bayesian integrated population model (IPM) to combine stage-structured information from multiple data sources and estimate demographic states and transitions (Schaub & Kéry, 2022). The data included classified counts from camera trap observations and census and proportion of calves estimates from aerial surveys.

A series of binomial distributions were used to inform both the counts of classified individuals each age class (calf, yearling, adult) and corresponding sex ratios from each camera trap observation. Using binomial distributions, as opposed to a multinomial distribution, allowed us to account for individuals that were classified by age, but not by sex. The probabilities for each of the binomial draws were informed by the estimated proportions of individuals in each age and sex class in the population on a given date, which was modeled using a Birth-Age-Survival (BAS) subprocess formulation of a population projection model (Newman et al., 2014).

For example, the count of calves at the ith camera trap observation was modeled as follows:

$$C_i \sim \text{Binomial}(N_i, p_{C_i})$$

where C_i is the number of calves, N_i is the total group size, and p_{C_i} is the sum of the expected proportions of male and female calves on the date of the ith observation.

The sex ratio of calves was modeled as follows:

$$F0_i \sim \text{Binomial}\left(F0_i + M0_i, \frac{p_{F0_i}}{p_{F0_i} + p_{M0_i}}\right)$$

where $F0_i$ is the number of female calves, $M0_i$ is the number of male calves, and p_{F0_i} and p_{M0_i} are the expected proportion of female and male calves on the date of the ith observation, respectively.

The model also estimated the fecundity rate, the proportion of fecund cows aged two and older, the annually-varying survival rates for each class, and produced derived estimates of several key population ratios (Fig. 2). The information from the aerial surveys was integrated into the model, helping inform the total number of individuals (i.e., the sum of the class-wise abundances) and the proportion of calves (i.e., $p_{F0_i} + p_{M0_i}$) on the dates of the aerial surveys. A Gaussian process regression (McElreath, 2016) accounted for the spatial and temporal correlation structure of the camera trap observations. Collectively, these methods could be applied non-invasively to a wide array of difficult to survey species to estimate key parameters that drive population dynamics.

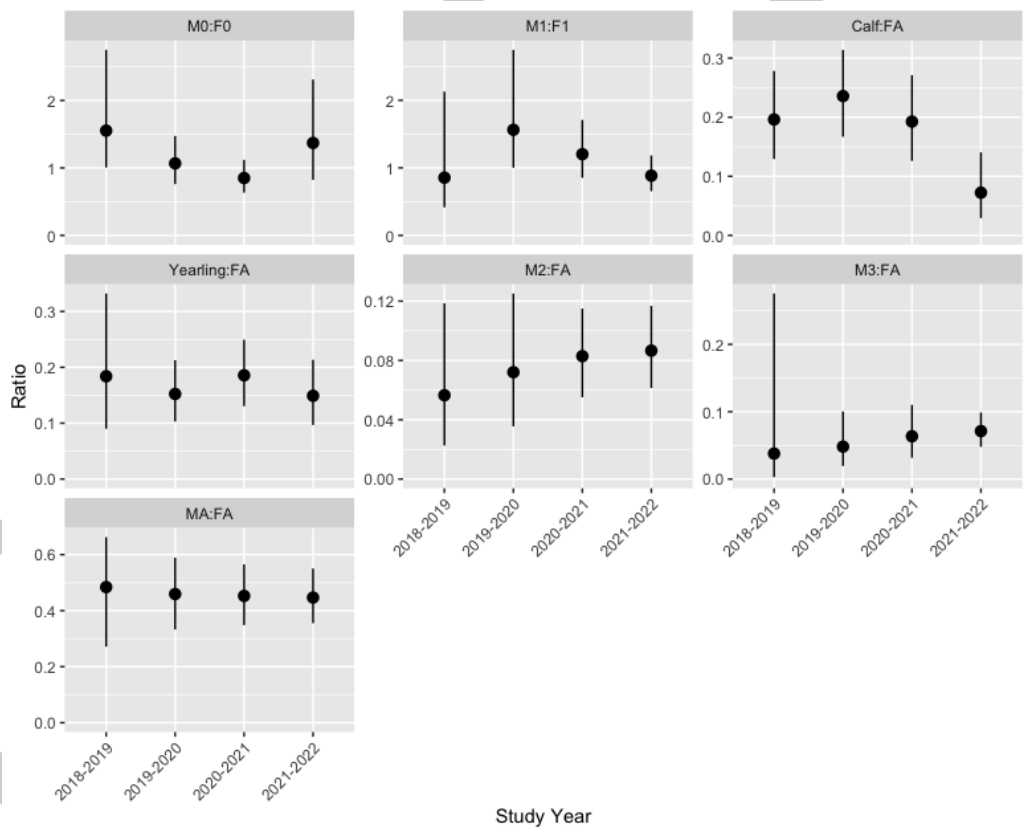

Figure 2: *Figure 2. Example of a prediction plot from the runbisonpic app, showing estimated population ratios for the Ronald Lake Wood Bison herd, by study year. M0 and F0 are male and female calves, M1 and F1 are male and female yearlings, Calf and Yearling represent all individuals within those age classes including those with unknown sex, M2 and M3 represent male two- and three-year-olds, MA represents males aged four and older, and FA represents females aged two and older.*

This method is implemented using four related R packages. The underlying functionality to check, clean, process, model, and visualize data is provided by `bisonpictools`. The other two R packages are apps that provide a user-friendly interface to `bisonpictools`. The first app is `shinybisonpic`, a web-based app that allows users to upload and explore data by viewing

camera locations and ratios of selected sex-age groups (e.g., Fig. 1). The second app is `runbisonpic`, a locally-run app that allows users with various skill levels to run the model to estimate the abundance by class, total abundance, survival and fecundity rates, and various sex-age ratios (e.g., Fig. 2). The `bisonpicsuite` package loads the three other packages.

Limitations

- The model can take over 5 hours to run.
- Key assumptions of the integrated population model include:
  - There is no grouping structure beyond what is accounted for by the covariance.
  - Every stage is equally detectable during a camera trap event.
  - Small and large groups are equally detectable.

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References

Bender, L. C. (2006). Uses of Herd Composition and Age Ratios in Ungulate Management. *Wildlife Society Bulletin*, *34*(4), 1225–1230. https://doi.org/10.2193/0091-7648(2006)34%5B1225:UOHCAA%5D2.0.CO;2

Buckland, S. T., Newman, K. B., Thomas, L., & Koesters, N. B. (2004). State-space models for the dynamics of wild animal populations. *Ecological Modelling*, *171*(1-2), 157–175. https://doi.org/10.1016/j.ecolmodel.2003.08.002

Caravaggi, A., Banks, P. B., Burton, A. C., Finlay, C. M. V., Haswell, P. M., Hayward, M. W., Rowcliffe, M. J., & Wood, M. D. (2017). A review of camera trapping for conservation behaviour research. *Remote Sensing in Ecology and Conservation*, *3*(3), 109–122. https://doi.org/10.1002/rse2.48

Chitwood, M. C., Lashley, M. A., Kilgo, J. C., Cherry, M. J., Conner, L. M., Vukovich, M., Ray, H. S., Ruth, C., Warren, R. J., DePerno, C. S., & Moorman, C. E. (2017). Are camera surveys useful for assessing recruitment in white-tailed deer? *Wildlife Biology*, *2017*(1), wlb.00178. https://doi.org/10.2981/wlb.00178

Fuller, J., Garrott, R., White, P. J., Aune, K., Roffe, T., & Rhyan, J. (2007). Reproduction and survival of Yellowstone bison. *The Journal of Wildlife Management*, *71*, 2365–2372. https://doi.org/10.2193/2006-201

Green, A. M., Chynoweth, M. W., & Şekercioğlu, Ç. H. (2020). Spatially explicit capture-recapture through camera trapping: A review of benchmark analyses for wildlife density estimation. *Frontiers in Ecology and Evolution*, *8*. https://doi.org/10.3389/fevo.2020.563477

Horne, J. S., & Thompson, S. J. (2021). Estimating survival of unmarked neonates with camera traps - Horne - 2021 - Ecosphere - Wiley Online Library. *Ecosphere*, *12(6)*, e03523. https://doi.org/10.1002/ecs2.3523

Ikeda, T., Takahashi, H., Tsuyoshi, Y., Igota, H., & Kaji, K. (2013). Evaluation of Camera Trap Surveys for Estimation of Sika Deer Herd Composition. *Mammal Study*, *38*(1), 29–33. https://doi.org/10.3106/041.038.0103

Laskin, D. N., Watt, D., Whittington, J., & Heuer, K. (2020). Designing a fence that enables free passage of wildlife while containing reintroduced bison: A multispecies evaluation.

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Wildlife Biology, *2020*(4), wlb.00751. https://doi.org/10.2981/wlb.00751

Magoun, A. J., Long, C. D., Schwartz, M. K., Pilgrim, K. L., Lowell, R. E., & Valkenburg, P. (2011). Integrating motion-detection cameras and hair snags for wolverine identification. *The Journal of Wildlife Management*, *75*(3), 731–739. https://doi.org/10.1002/jwmg.107

McElreath, R. (2016). *Statistical Rethinking: A Bayesian course with examples in R and Stan*. CRC Press/Taylor & Francis Group.

Mizuki, I., Itô, H., Yamasaki, M., Fukumoto, S., Okamoto, Y., Katsuki, M., Fukushima, K., Sakai, M., Sakaguchi, S., Fujiki, D., Nakagawa, H., Ishihara, M. I., & Takayanagi, A. (2020). Seasonal and annual fluctuations of deer populations estimated by a Bayesian state–space model. *PLOS ONE*, *15*(6), e0225872. https://doi.org/10.1371/journal.pone.0225872

Nakashima, Y., Hongo, S., & Akomo-Okoue, E. F. (2020). Landscape-scale estimation of forest ungulate density and biomass using camera traps: Applying the REST model. *Biological Conservation*, *241*, 108381. https://doi.org/10.1016/j.biocon.2019.108381

Newman, K. B., Buckland, S. T., Morgan, B. J. T., King, R., Borchers, D. L., Cole, D. J., Besbeas, P., Gimenez, O., & Thomas, L. (2014). *Modelling population dynamics: Model formulation, fitting and assessment using state-space methods*. ISBN: 978-1-4939-0976-6

Newman, K. B., King, R., Elvira, V., De Valpine, P., McCrea, R. S., & Morgan, B. J. T. (2023). State-space models for ecological time-series data: Practical model-fitting. *Methods in Ecology and Evolution*, *14*(1), 26–42. https://doi.org/10.1111/2041-210X.13833

Paterson, J. T., Proffitt, K., Rotella, J., & Garrott, R. (2019). An improved understanding of ungulate population dynamics using count data: Insights from western Montana. *PLOS ONE*, *14*(12), e0226492. https://doi.org/10.1371/journal.pone.0226492

Schaub, M., & Kéry, M. (2022). *Integrated population models: Theory and ecological applications with R and JAGS*. Academic Press. ISBN: 978-0-323-90810-8

Singh, H., Sharief, A., Joshi, B. D., Kumar, V., Mukherjee, T., Chandra, K., Bhardwaj, N., Thakur, M., & Sharma, L. K. (2022). Multi-species occupancy modeling suggests interspecific interaction among the three ungulate species. *Scientific Reports*, *12*(1), 17602. https://doi.org/10.1038/s41598-022-20953-7

Steenweg, R., Whittington, J., Hebblewhite, M., Forshner, A., Johnston, B., Petersen, D., Shepherd, B., & Lukacs, P. M. (2016). Camera-based occupancy monitoring at large scales: Power to detect trends in grizzly bears across the Canadian Rockies. *Biological Conservation*, *201*, 192–200. https://doi.org/10.1016/j.biocon.2016.06.020

Wittmer, H., Mclellan, B., Seip, D., Young, J., Kinley, T., Watts, G., & Hamilton, D. (2005). Population dynamics of the endangered mountain ecotype of woodland caribou (Rangifer tarandus caribou) in British Columbia, Canada. *Canadian Journal of Zoology*, *83*, 407–418. https://doi.org/10.1139/z05-034

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